

Improving PAC-Bayes bounds for neural nets

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Introduction

Recent works like Dziugaite and Roy 2017 and Zhou et al. 2018 have shown that one can obtain non-vacuous “computational” PAC-Bayesian bounds on the risk of neural nets. In this work we improve on existing theoretical PAC-Bayesian risk bounds by using data-dependent priors sensitive to the geometrical properties of training and by considering carefully chosen clusters of multiple nets in tandem.

Notation

$\delta f(x)$ is the set of **subgradients** of the point $x \in \mathbb{R}^n$. Then the subgradient $g \in \delta f(x)$ satisfies $f(x) - f(y) \leq g^\top(x - y)$. $e_1, \dots, e_n$ is the canonical basis of $\mathbb{R}^n$.

Fastest theoretical speed for Gradient Descent

Theorem (1)

Let $t \leq \frac{n-1}{2}$ and $\beta > 0$. There exists a β -smooth convex function f such that for any black-box procedure satisfying $x_{t+1} \in \text{Span}(g_1, \dots, g_t)$ with $g_s \in \delta f(x_s)$:

$$\min_{1 \leq s \leq t} f(x_s) - f(x^*) \geq \frac{3\beta \|x_1 - x^*\|^2}{32 (t+1)^2}$$

This gives us a bound on the speed of gradient descent of $O(1/t^2)$. We observe that this bound is achieved by Nesterov’s Accelerated Gradient Descent

Crafting a β -smooth function f

To construct a suitable function f that satisfies Theorem (1), Bubeck constructs a symmetric and tri-diagonal matrix

$$(A_k)_{i,j} = \begin{cases} 2, & i = j, \ i \leq k \\ -1, & j \in \{i-1, i+1\}, \ i \leq k, \ j \neq k+1 \\ 0, & \text{otherwise} \end{cases}$$

and defines f as:

$$f(x) = \frac{\beta}{8} x^\top A_{2t+1} x - \frac{\beta}{4} x^\top e_1$$

Descent Lemma

The following lemma establishes the per-step bound for gradient descent on smooth functions. The original paper states the constrained case but for Nesterov’s accelerated gradient descent in the smooth function case, we only need the unconstrained version.

Lemma (2)

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a convex and β -smooth function. Let $x, y \in \mathbb{R}^n$ and $x^+ = x - \frac{1}{\beta} \nabla f(x)$. Then the following holds true:

$$f(x^+) - f(y) \leq \beta(x^+ - x)^\top(x - y) - \frac{1}{2\beta} \|\beta(x^+ - x)\|^2$$

Nesterov’s accelerated gradient descent

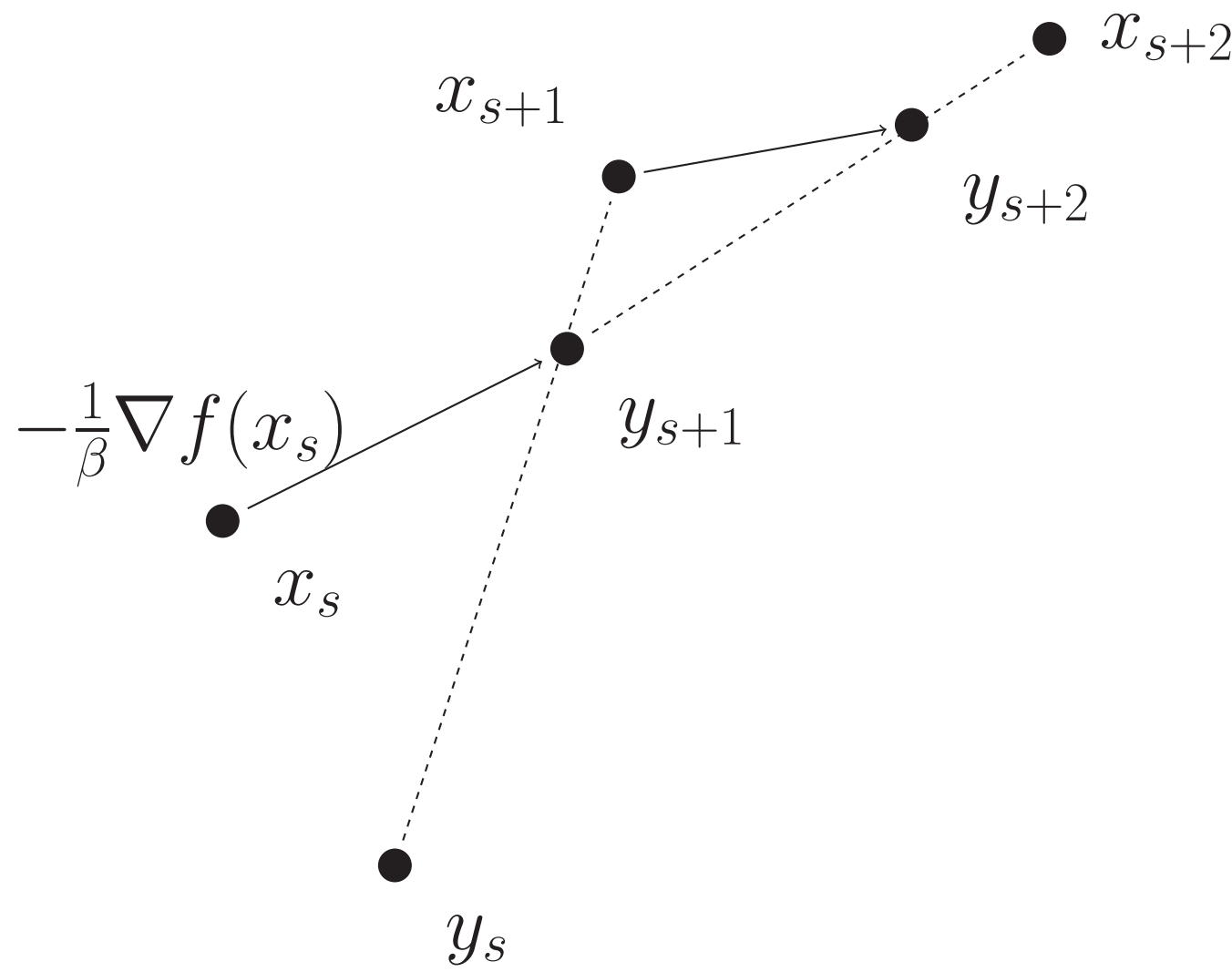


Figure: Illustration of Nesterov’s accelerated gradient descent.

First we introduce the following sequences:

$$\lambda_0 = 0, \ \lambda_t = \frac{1 + \sqrt{1 + 4\lambda_{t-1}^2}}{2}, \text{ and } \gamma_t = \frac{1 - \lambda_t}{\lambda_{t+1}}$$

Nesterov’s accelerated gradient descent is described as follows. Start at an arbitrary initial point $x_1 = y_1$ and then iterate the following equations for $t \geq 1$

$$y_{t+1} = x_t - \frac{1}{\beta} \nabla f(x_t)$$

$$x_{t+1} = (1 - \gamma_t)y_{t+1} - \gamma_t y_t$$

Theorem (3)

Let f be a convex and β -smooth function, then Nesterov’s accelerated gradient descent satisfies

$$f(y_t) - f(x^*) \leq \frac{2\beta \|x_1 - x^*\|^2}{t^2}$$

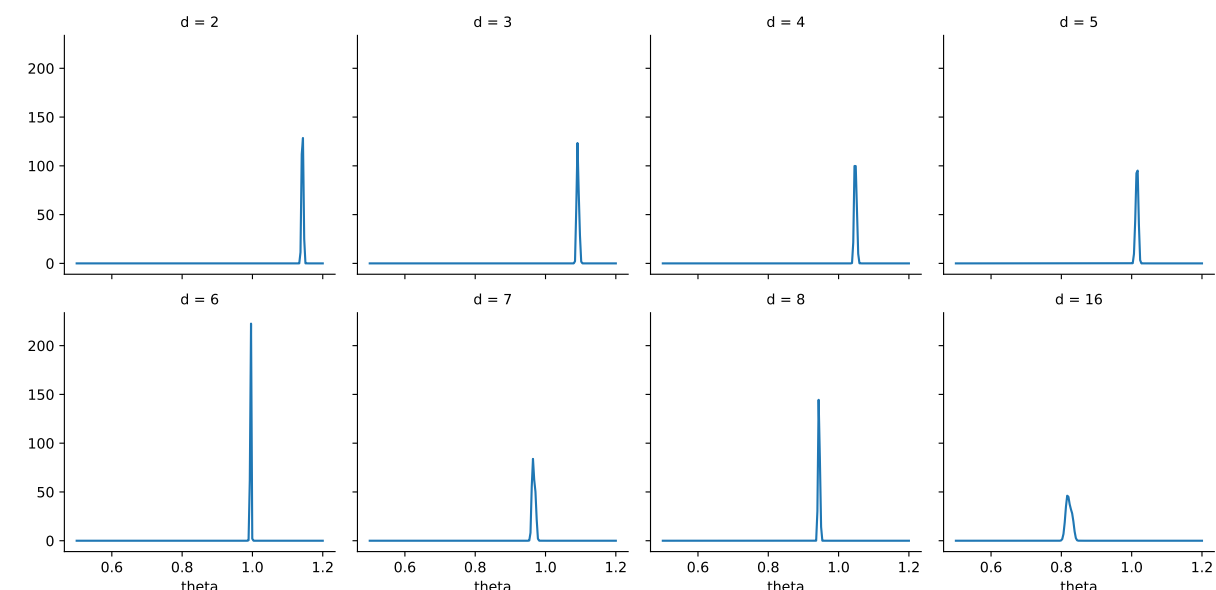
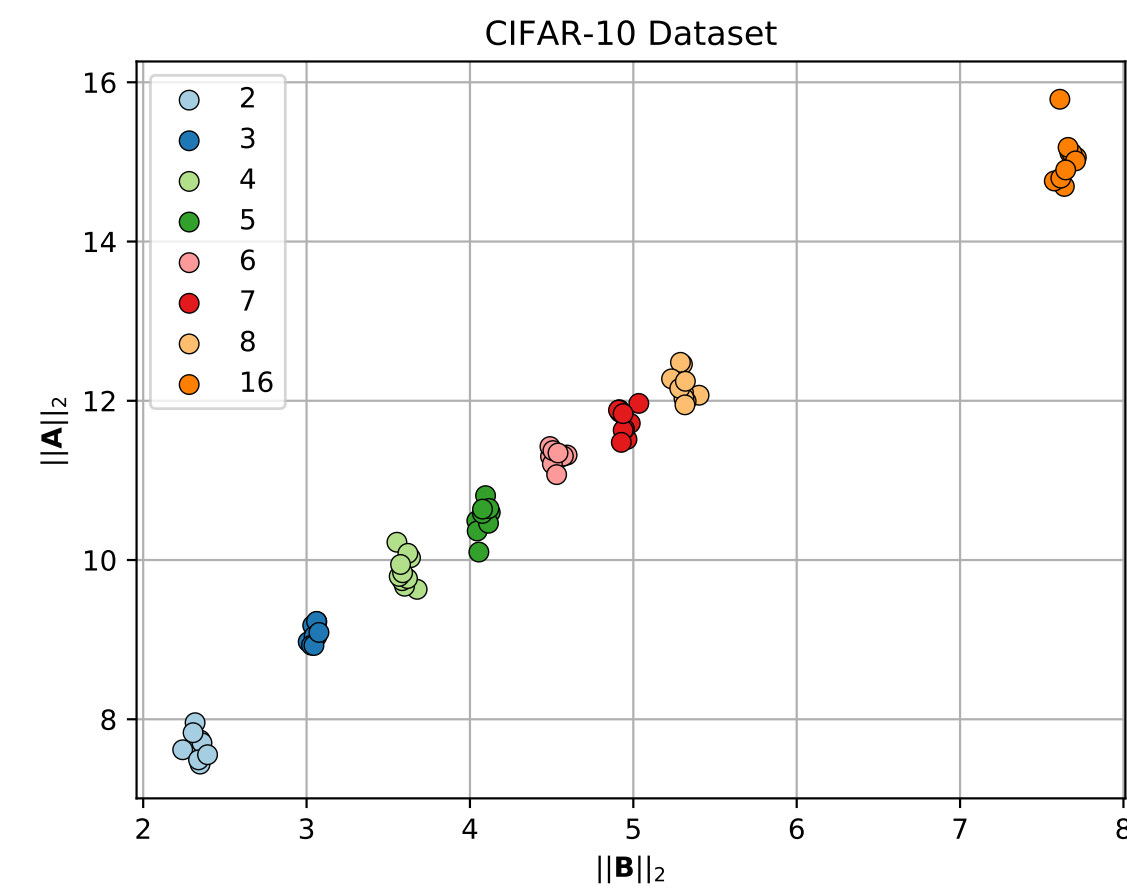
This gives us the $O(1/t^2)$ complexity. Write more about it here

Main Ideas for Theorem 3

- Nesterov’s accelerated gradient descent uses the ideas from classical momentum gradient descent (Sutskever et al., 2013). x_{t+1} can be rewritten as $x_{t+1} = y_{t+1} + \gamma_t(y_{t+1} - y_t)$ which reads as x_{t+1} overshoots y_{t+1} along the vector $(y_{t+1} - y_t)$ with some “momentum” given by γ_t
- The choices for λ_t and γ_t are not arbitrary.

Demonstration of stability of the relevant geometrical properties of training

In the *figure to the left* we plot the initial parameter norm $\|B\|_2$ vs the final parameter norm $\|A\|_2$ for training nets of different depths at width 100 on the CIFAR-10 dataset. Thus its demonstrated that the multiplicative factor with which the norm increases during training is fairly stable to architectural changes.



In the *figure to the right* is displayed the Gaussian kernel density estimate on 10 trials of the experiment (at different depths and 100 width) measuring the angular deviation of the weight vector under training. This deflection angle only decreases slightly with depth and showed negligible dependence over the widths tried.

References

- Dziugaite, Gintare Karolina and Daniel M Roy (2017). “Computing nonvacuous generalization bounds for deep (stochastic) neural networks with many more parameters than training data”. In: *arXiv preprint arXiv:1703.11008*.
- Zhou, Wenda et al. (2018). “Non-vacuous generalization bounds at the imagenet scale: a PAC-bayesian compression approach”. In: *arXiv preprint arXiv:1804.05862*.